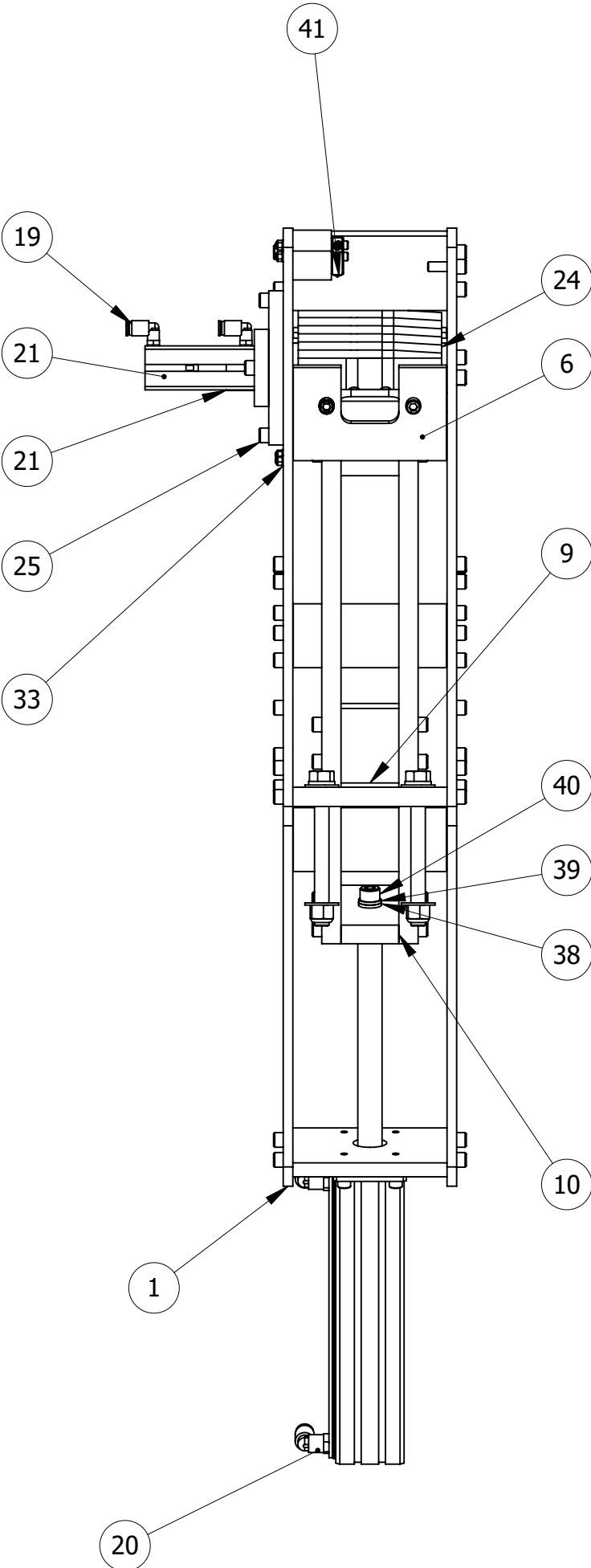
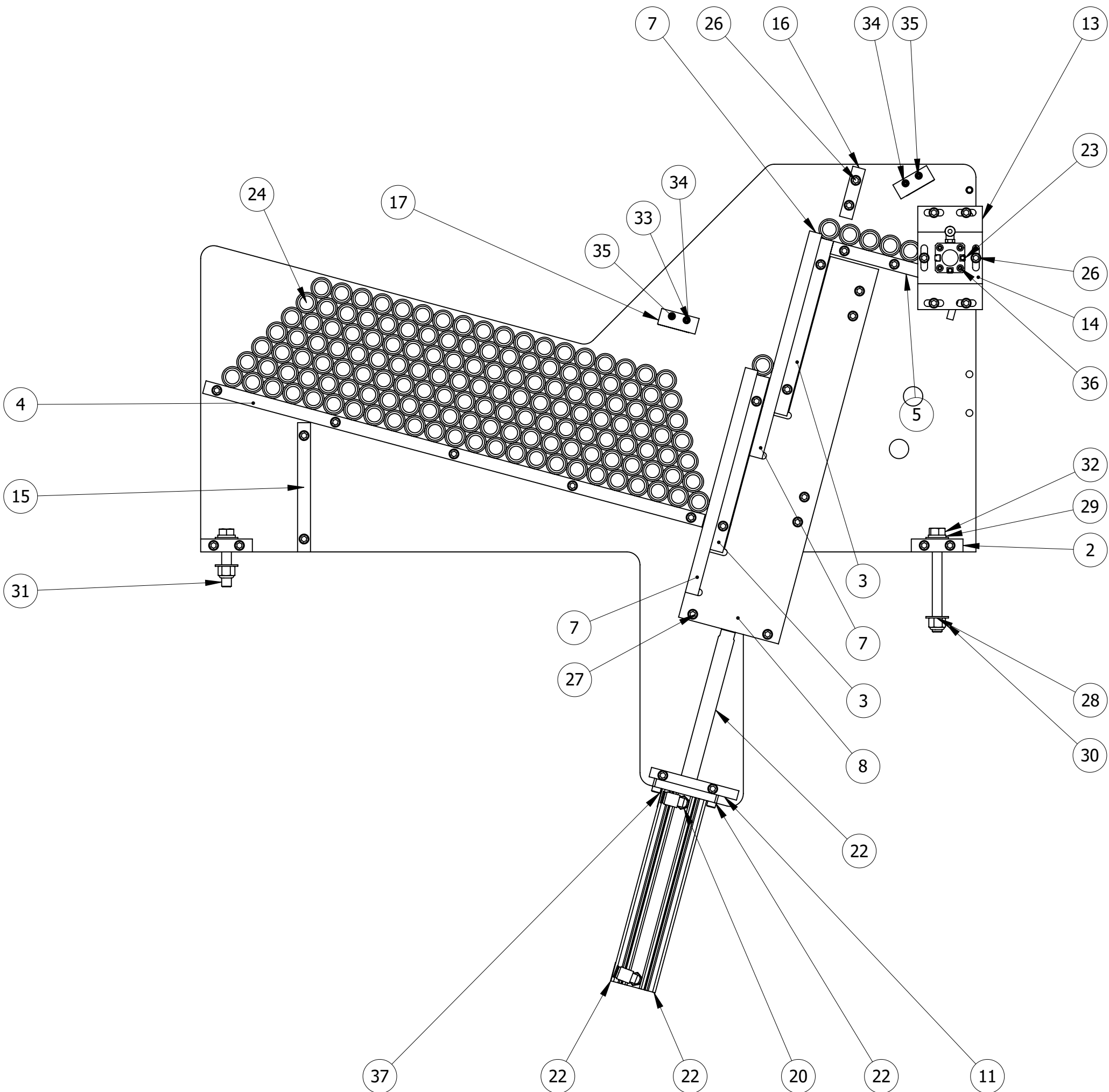
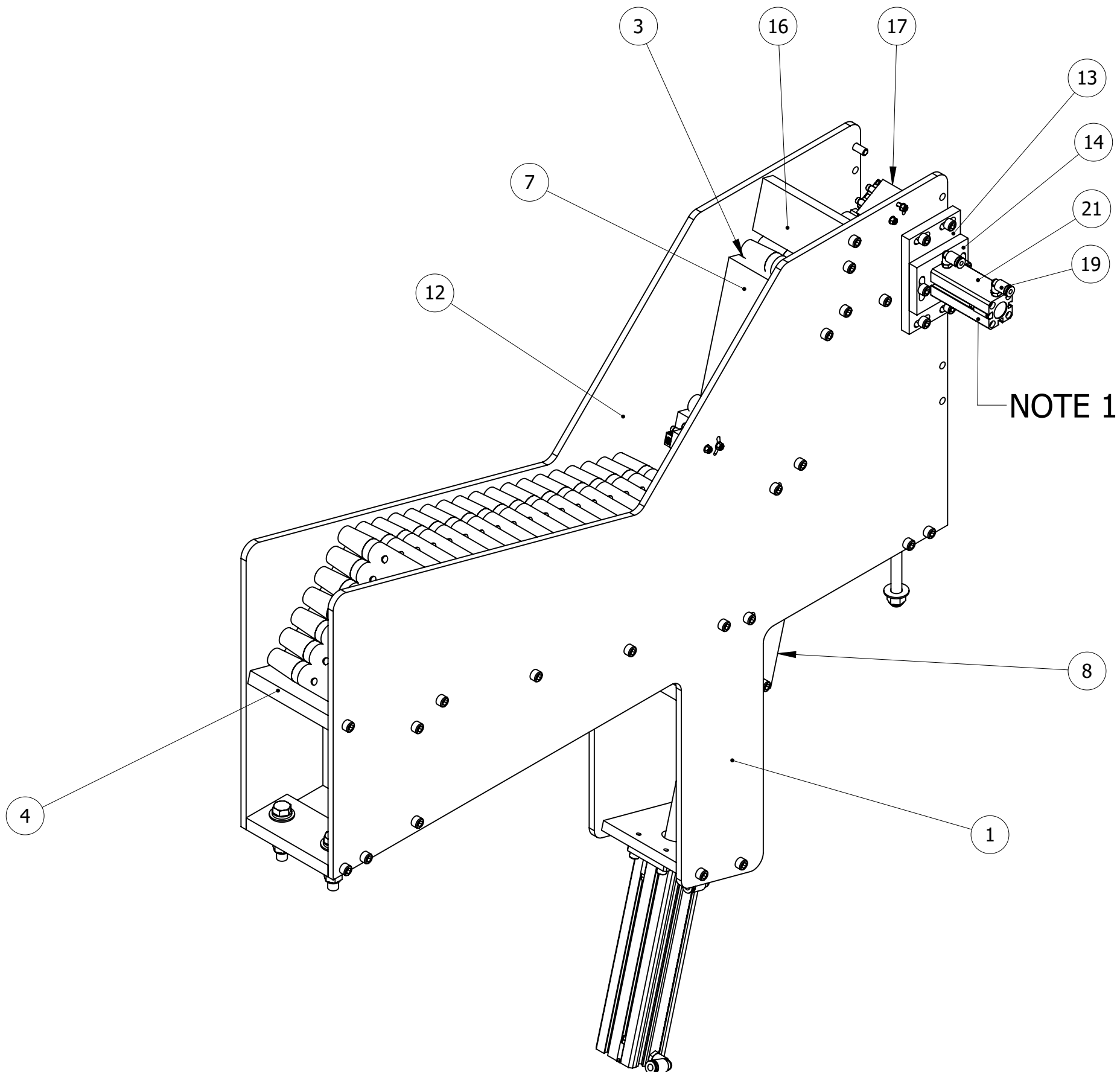


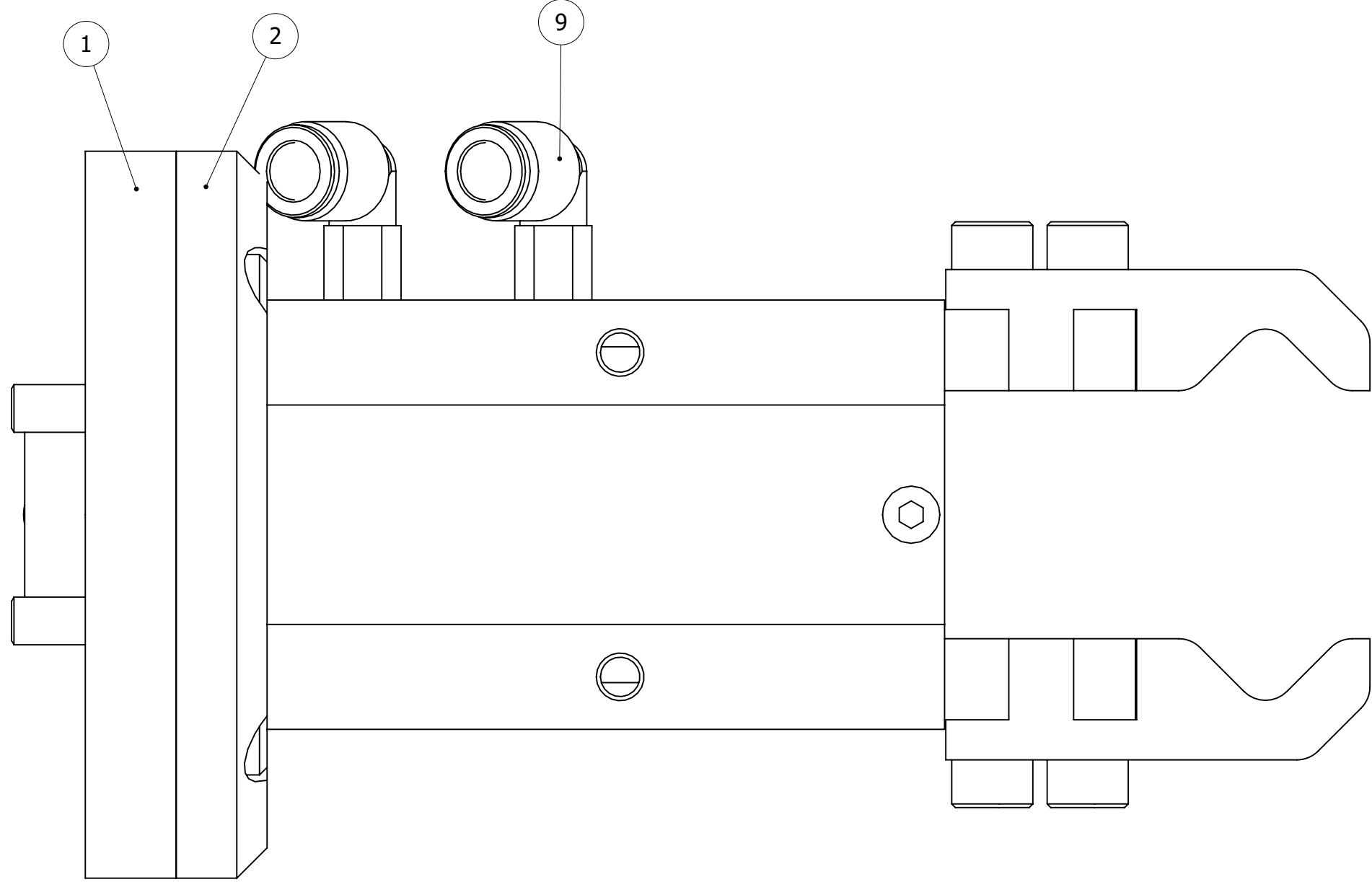
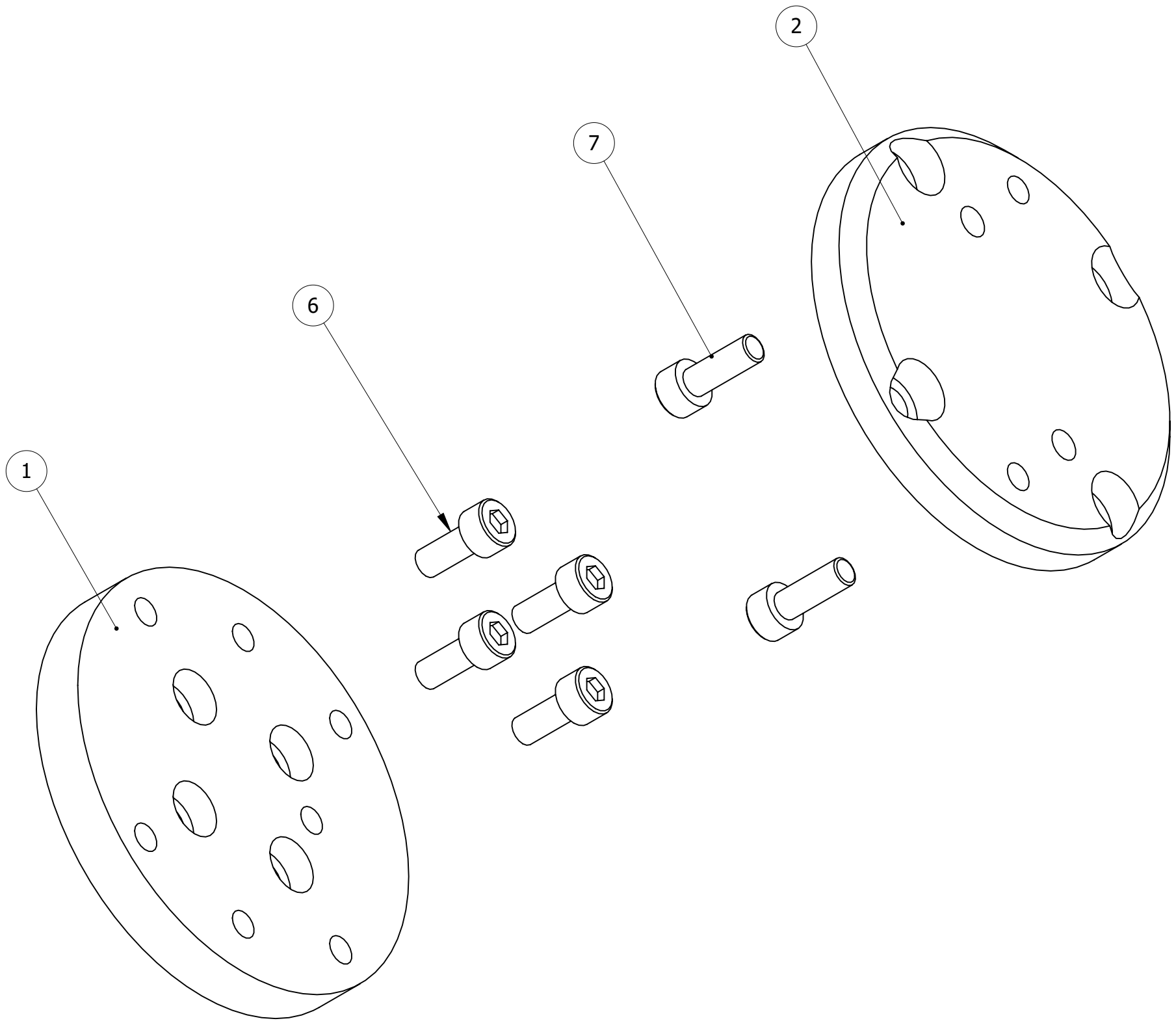
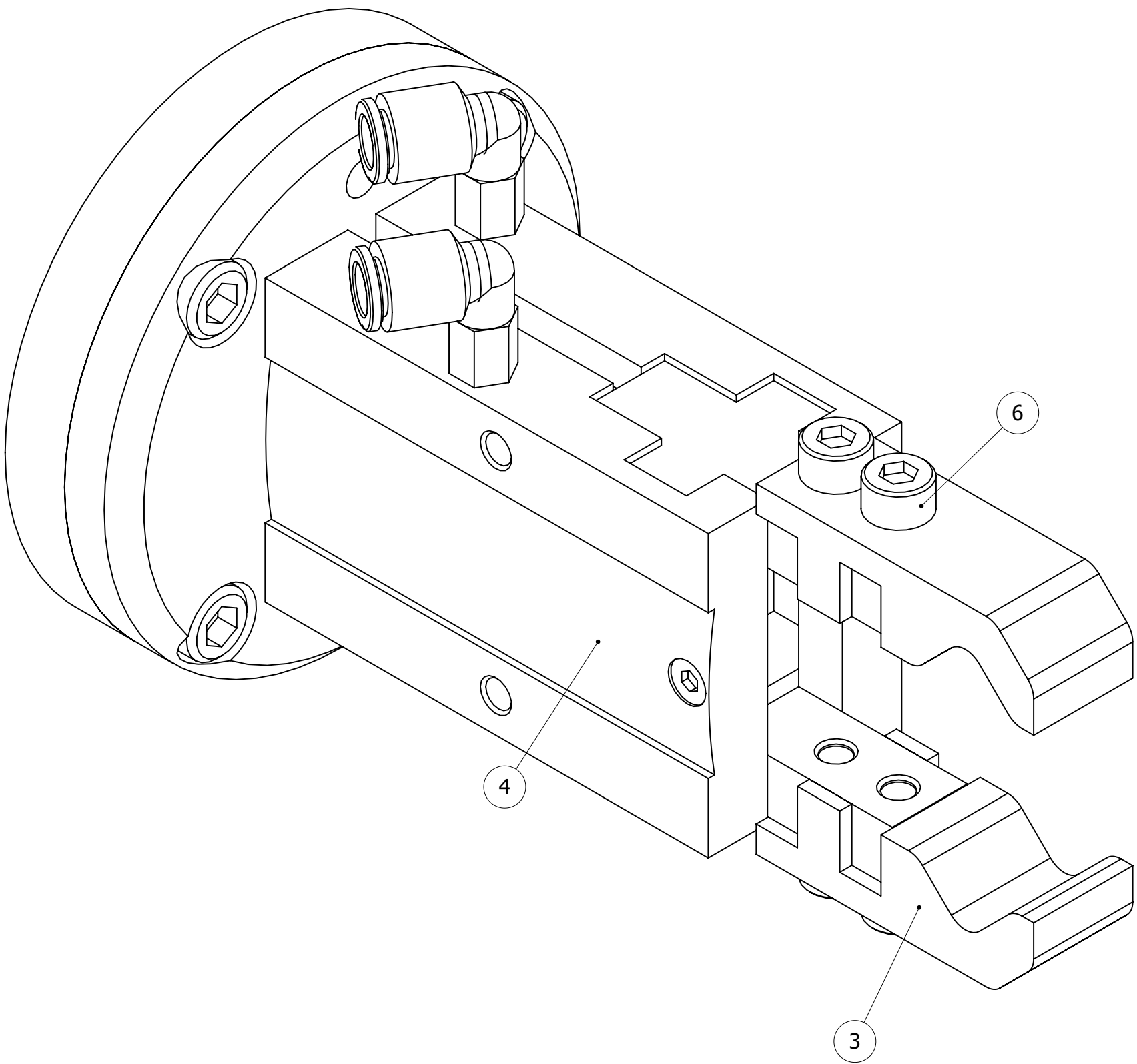
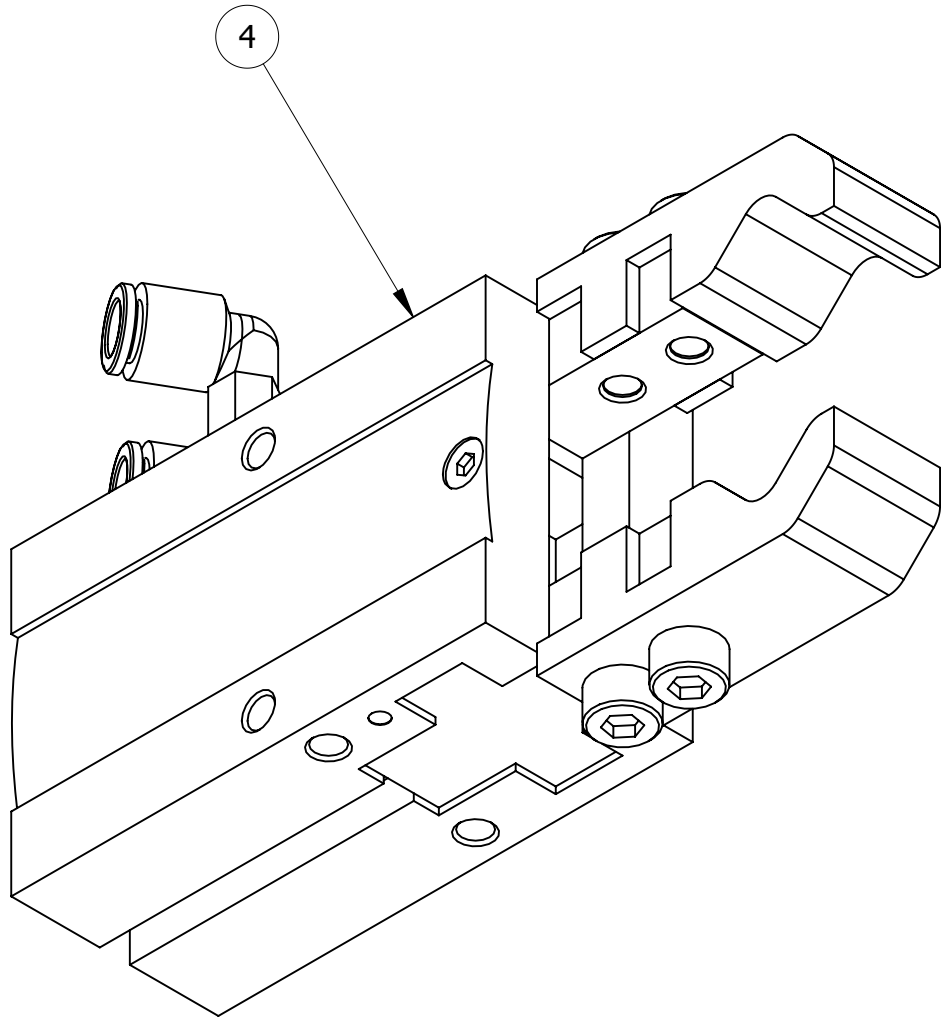
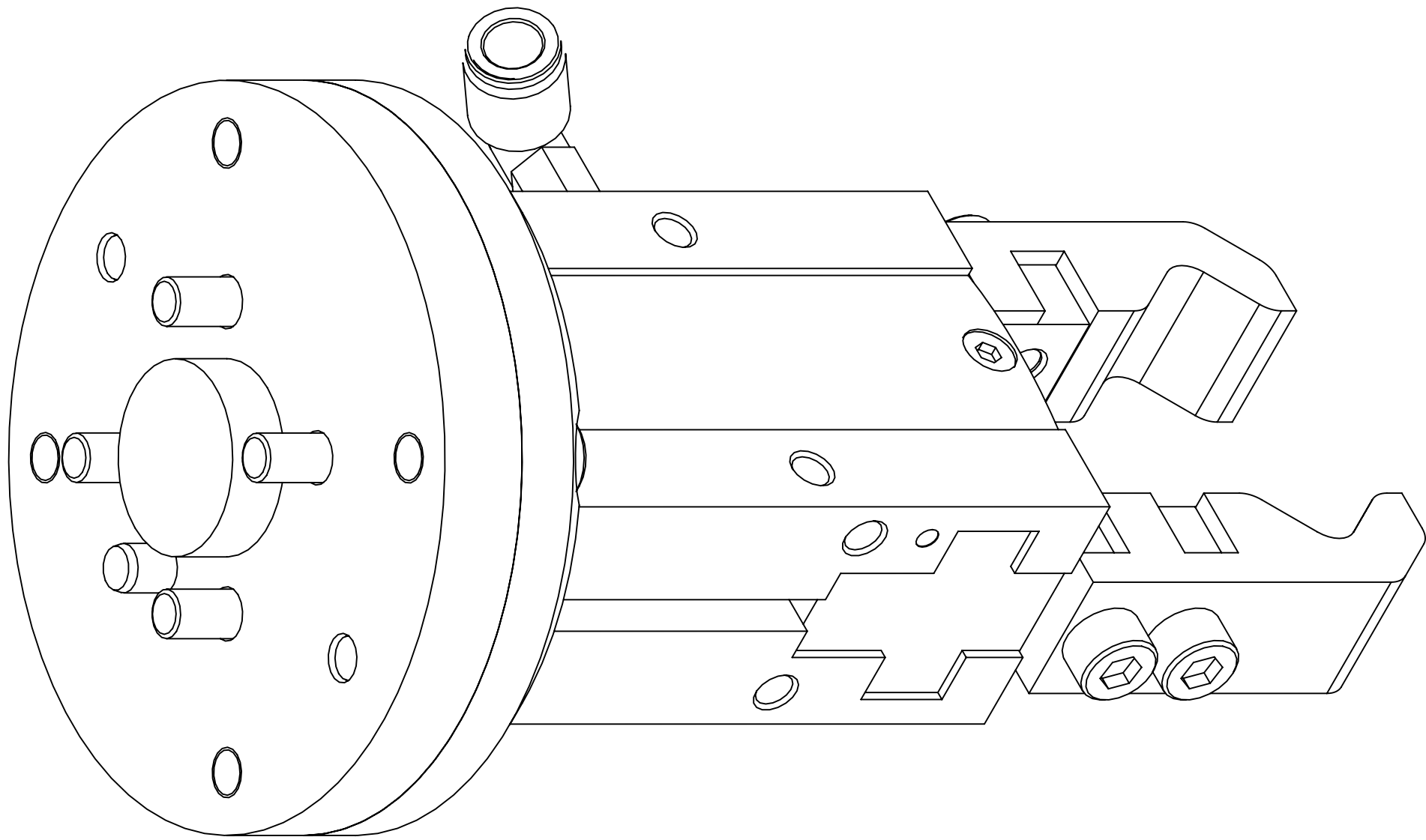
ITEM NO.	QTY.	PART NUMBER	REV	DESCRIPTION	MATERIAL	FINISH	NOTES	SOURCE
1	1	0118 02 2001 External Step Feeder Side			Aluminum 6061 Alloy	Anodize Clear	Machined Part	
2	2	0118 02 2002 Step Feeder Base			Aluminum 6061 Alloy	Anodize Clear	Machined Part	
3	2	0118 02 2003 Middle Step			Aluminum 6061 Alloy	Anodize Clear	Machined Part	
4	1	0118 02 2004 Step Feeder Catch			Aluminum 6061 Alloy	Anodize Clear	Machined Part	
5	1	0118 02 2005 Step Feeder Accumulator			Aluminum 6061 Alloy	Anodize Clear	Machined Part	
6	1	0118 02 2006 Stop Block			Aluminum 6061 Alloy	Anodize Clear	Machined Part	
7	2	0118 02 2007 Step Block			HDPE (black)	Natural	Machined Part	
8	2	0118 02 2008 Step Bracket			Aluminum 6061 Alloy	Anodize Clear	Machined Part	
9	2	0118 02 2009 Bracket Brace			Aluminum 6061 Alloy	Anodize Clear	Machined Part	
10	1	0118 02 2010 Cylinder Bar			Aluminum 6061 Alloy	Anodize Clear	Machined Part	
11	1	0118 02 2011 Cylinder Mount			Aluminum 6061 Alloy	Anodize Clear	Machined Part	
12	1	0118 02 2012 Internal Step Feeder Side			Aluminum 6061 Alloy	Anodize Clear	Machined Part	
13	1	0118 02 2013 Horiz Cylinder Adjust Plate			Aluminum 6061 Alloy	Anodize Clear	Machined Part	
14	1	0118 02 2014 Vert Cylinder Adjust Plate			Aluminum 6061 Alloy	Anodize Clear	Machined Part	
15	1	0118 02 2015 Cover Plate			Aluminum 6061 Alloy	Anodize Clear	Machined Part	
16	1	0118 02 2016 Singulator Plate			Aluminum 6061 Alloy	Anodize Clear	Machined Part	
17	2	0118 02 2017 Sensor Spacer			Aluminum 6061 Alloy	Anodize Clear	Machined Part	
18	1	0118 02 2201 Contactor Inspection Cap			4140 Steel	Zinc Plated	Machined Part	
19	2	CKD~GWL4-M5		4mm Elbow, M5 Thread	As Purchased	As Purchased	Purchased Part	
20	2	CKD~GWL6-6		6mm Elbow, 1/8	As Purchased	As Purchased	Purchased Part	
21	1	CKD~SSD2-L-16-40-T3PH-D		16mm Bore x 40mm Stroke Cylinder	As Purchased	As Purchased	Purchased Part	
22	1	CKD~SSD2-L-32-150-T3PH-D-FA		32mm Bore x 150mm Stroke w 2 Switches and Flange	As Purchased	As Purchased	Purchased Part	
23	1	CKD~T3PH		3 Wire 1 Color PNP Cylinder Proxy Sensor	As Purchased	As Purchased	Purchased Part	
24	155	ABB Female Contact			Beryllium Copper, UNS C17000	Natural	Customer Part	
25	4	.25-20 x 0.63 SHCS ZC		1/4-20 x .63 SHCS Zinc Plated	Alloy Steel	Zinc Plated	Fastener	
26	54	.25-20 x 0.75 SHCS ZC		1/4-20 x .75 SHCS Zinc Plated	Alloy Steel	Zinc Plated	Fastener	
27	12	.25-20 x 1.00 SHCS ZC		.25-20 x 1.00 SHCS	Alloy Steel	Zinc Plated	Fastener	
28	8	.38 Flat Washer ZC		3/8 Flat Washer ZC	Alloy Steel	Zinc Plated	Fastener	
29	4	.38 Lock Washer ZC		3/8 Lock Washer Zinc Plated	Alloy Steel	Zinc Plated	Fastener	
30	4	.38-16 Nylon Lock Nut ZC		3/8-16 Nylon Lock Nut ZC	Alloy Steel	Zinc Plated	Fastener	
31	2	.38-16 x 2.00 HHCS ZYC		3/8-16 x 2.00 HHCS	Alloy Steel	Zinc Yellow-Chromate Plated	Fastener	
32	2	.38-16 x 3.75 HHCS ZYC		3/8-16 x 3.75 Hex Head	Alloy Steel	Zinc Yellow-Chromate Plated	Fastener	
33	4	M3 Flat Washer ZC		M3 Flat Washer	Alloy Steel	Zinc Plated	Fastener	
34	4	M3X0.5 Nylon Lock Nut ZC		M3X0.5 Nylon Lock Nut	Steel	Zinc Plated	Fastener	
35	4	M3X0.5 x 45 SHCS BO		M3X0.5 x 35 SHCS	Alloy Steel	Black Oxide	Fastener	
36	4	M4X0.7 x 12 SHCS ZC		M4X0.7 x 12 SHCS Zinc Plated	Alloy Steel	Zinc Plated	Fastener	
37	5	M5X0.8 x 16 SHCS ZC		M5X0.8 x 16 SHCS	Alloy Steel	Zinc Plated	Fastener	
38	1	M8 Flat Washer ZC		M8 Flat Washer	Alloy Steel	Zinc Plated	Fastener	
39	1	M8 Lock Washer ZC		M8 Lock Washer Zinc Plated	Alloy Steel	Zinc Plated	Fastener	
40	1	M8X1.25 x 30 SHCS ZC		M8X1.25 x 30 SHCS	Alloy Steel	Zinc Plated	Fastener	
41	2	Banner~Q2XABAF150-Q		Photoelectric Sensor 150mm Range 4 Pin M8 Connector	As Purchased	As Purchased	Purchased Part	



Note 1:
Piston has 3 sensors

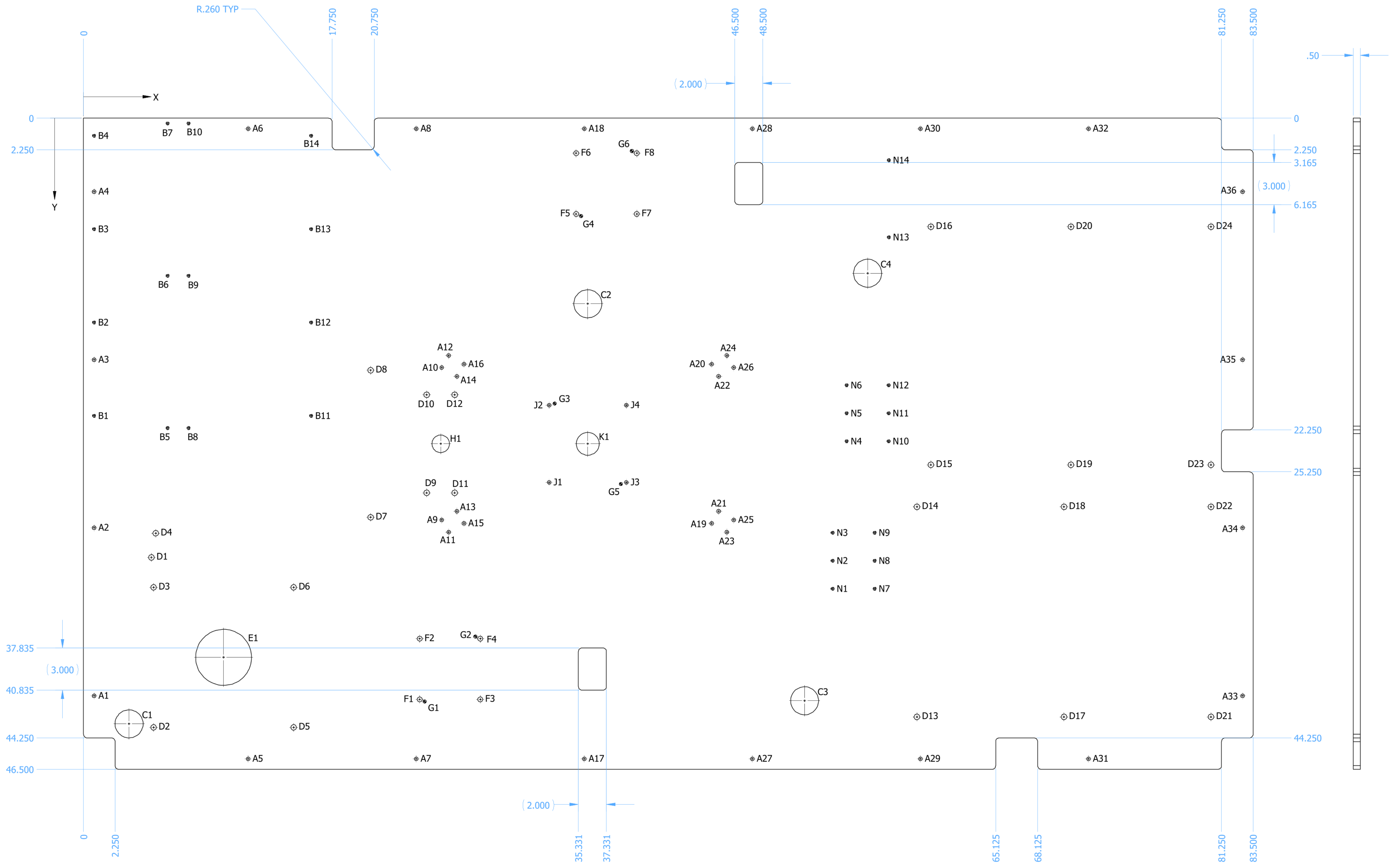
MACHINED BY	DATE / TIME	QTY / THIS ORDER - 1
INSPECTED BY	DATE / TIME	QTY / ASSY - 1
		MATERIAL -
		FINISH - NA
		NOTES - Assembly
		UNLESS OTHERWISE SPECIFIED ALL DIMENSIONS ARE IN INCHES
		TOLERANCES
		DECIMALS .XX ±.01 .XXX ±.005 .XXXX ±.0005
		ANGLES ±1° .X ±.5° .XX ±.1°
		FRACTIONS ±1/16

ITEM NO.	QTY.	PART NUMBER	REV	DESCRIPTION	MATERIAL	FINISH	NOTES	SOURCE
1	1	0118 02 3502 Robot EOAT Plate			Aluminum 6061 Alloy	Anodize Clear	Machined Part	
2	1	0118 02 3503 Parallel Gripper Plate			Aluminum 6061 Alloy	Anodize Clear	Machined Part	
3	2	0118 02 3504 Parallel Gripper Finger			Aluminum 6061 Alloy	Anodize Clear	Machined Part	
4	1	CKD-FH126-D		Pneumatic Parallel Gripper		As Purchased	Purchased Part	
5	4	.25-20 x 0.50 SHCS ZC		1/4-20 x .50 SHCS	Alloy Steel	Zinc Plated	Fastener	
6	8	M5X0.8 x 12 SHCS ZC		M5X0.8 x 12 SHCS	Alloy Steel	Zinc Plated	Fastener	
7	2	M5X0.8 x 14 SHCS ZC		M5X0.8 x 14 SHCS	Steel	Zinc Plated	Fastener	
8	3	McMaster-Carr~91595A352		Dowel Pin; 5mm OD x 16mm long	Alloy Steel	Natural	Purchased Part	
9	2	SMC~KQ2L06-M5A		6mm x M5 One Touch Fitting	Bronze SAE 510	As Purchased	Purchased Part	



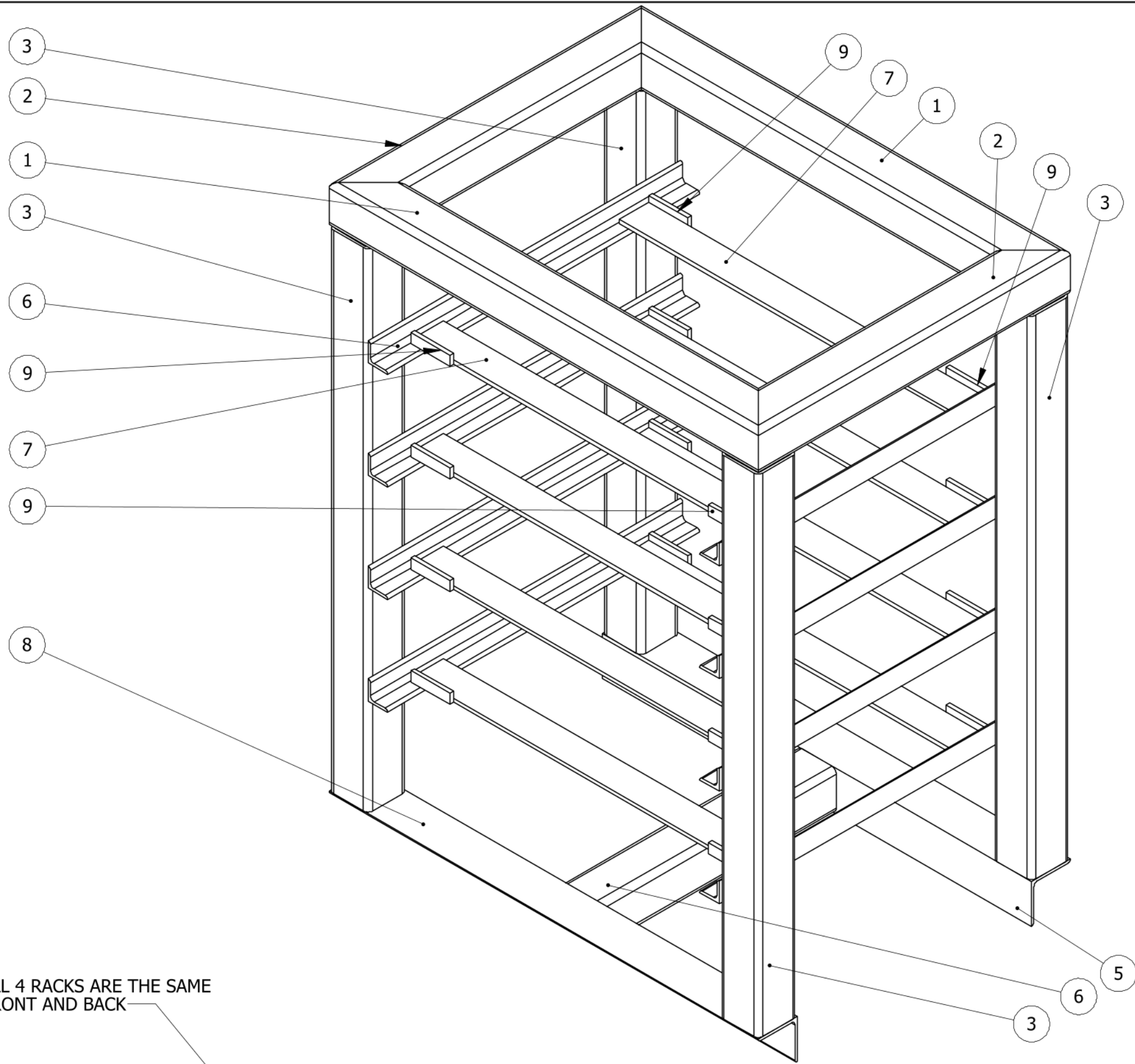
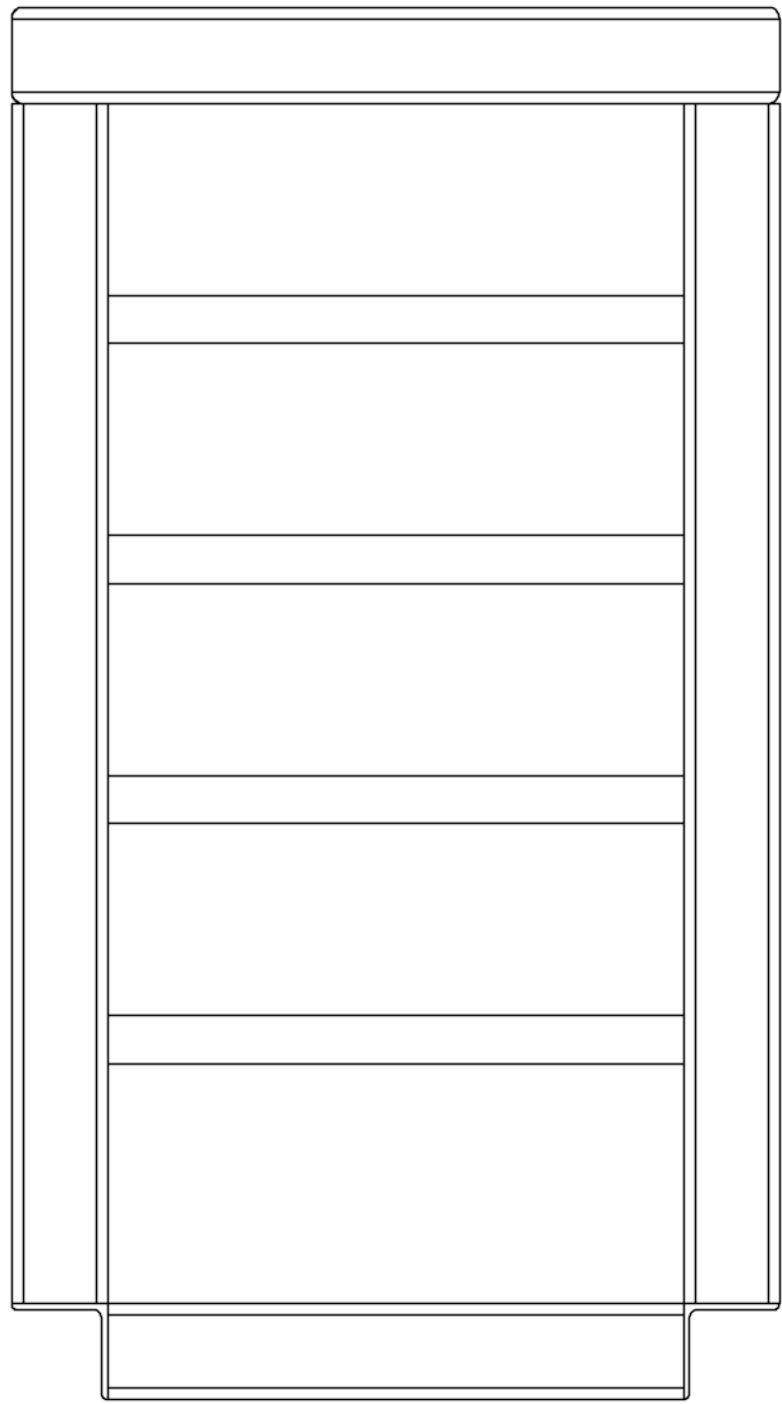
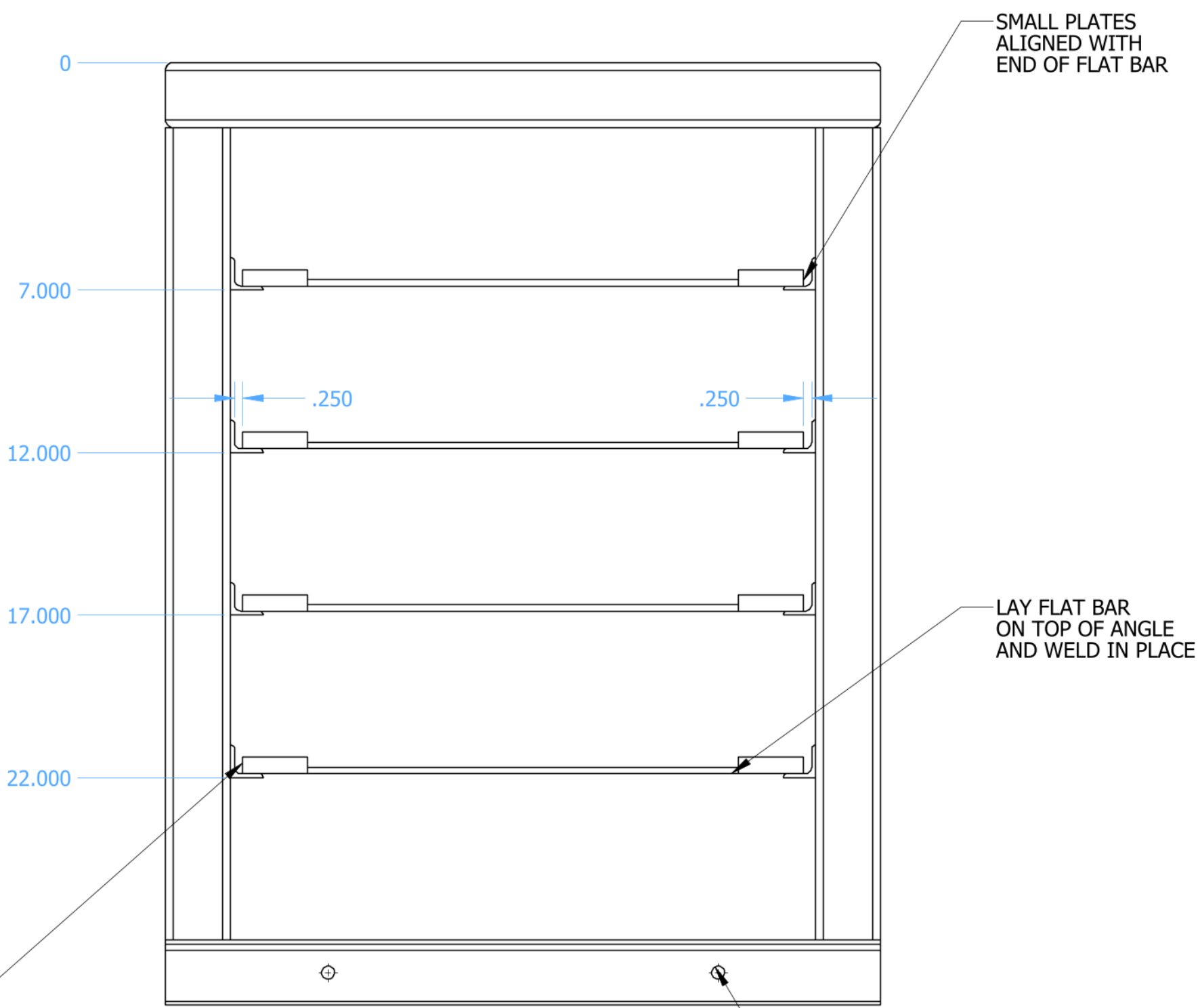
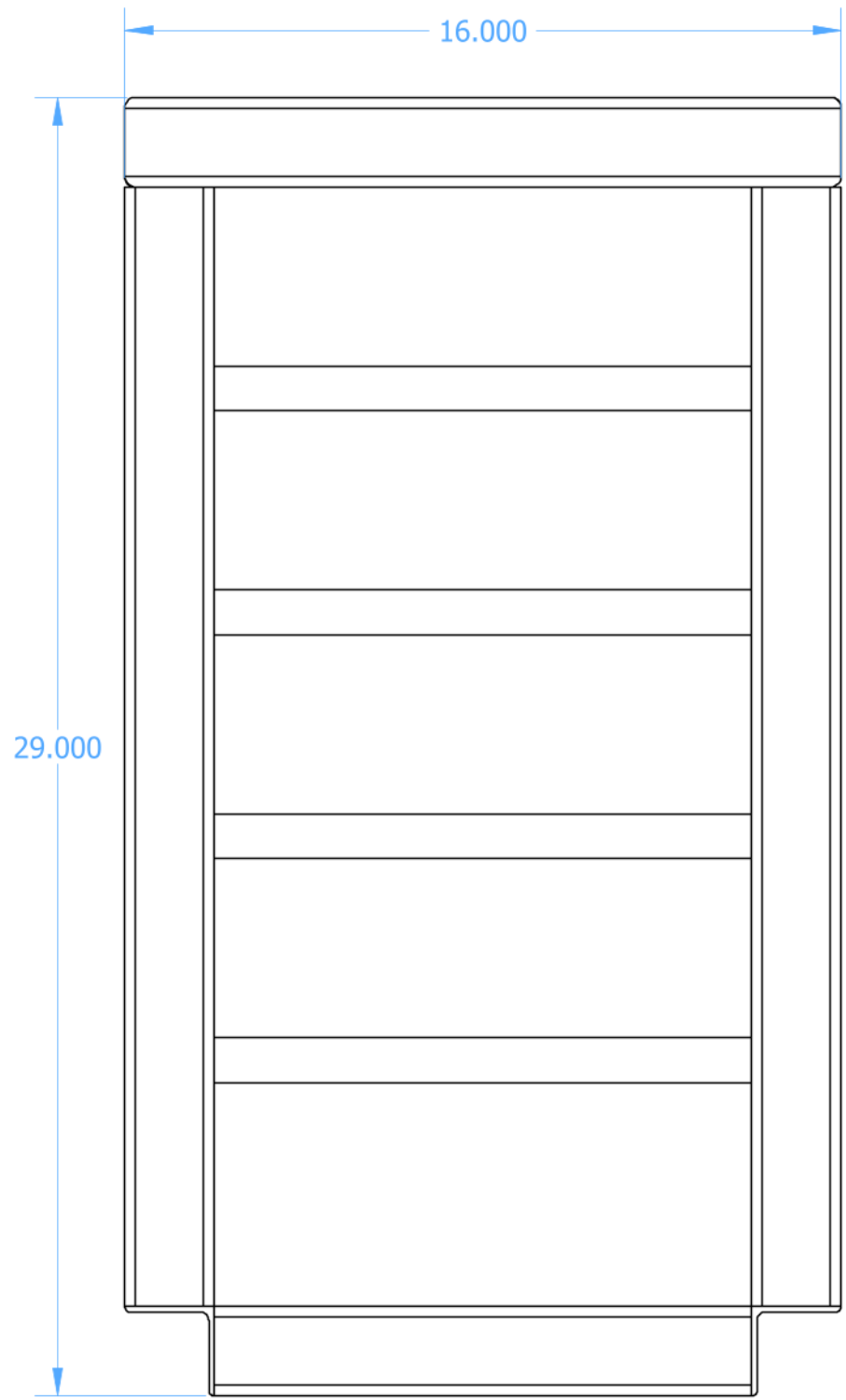
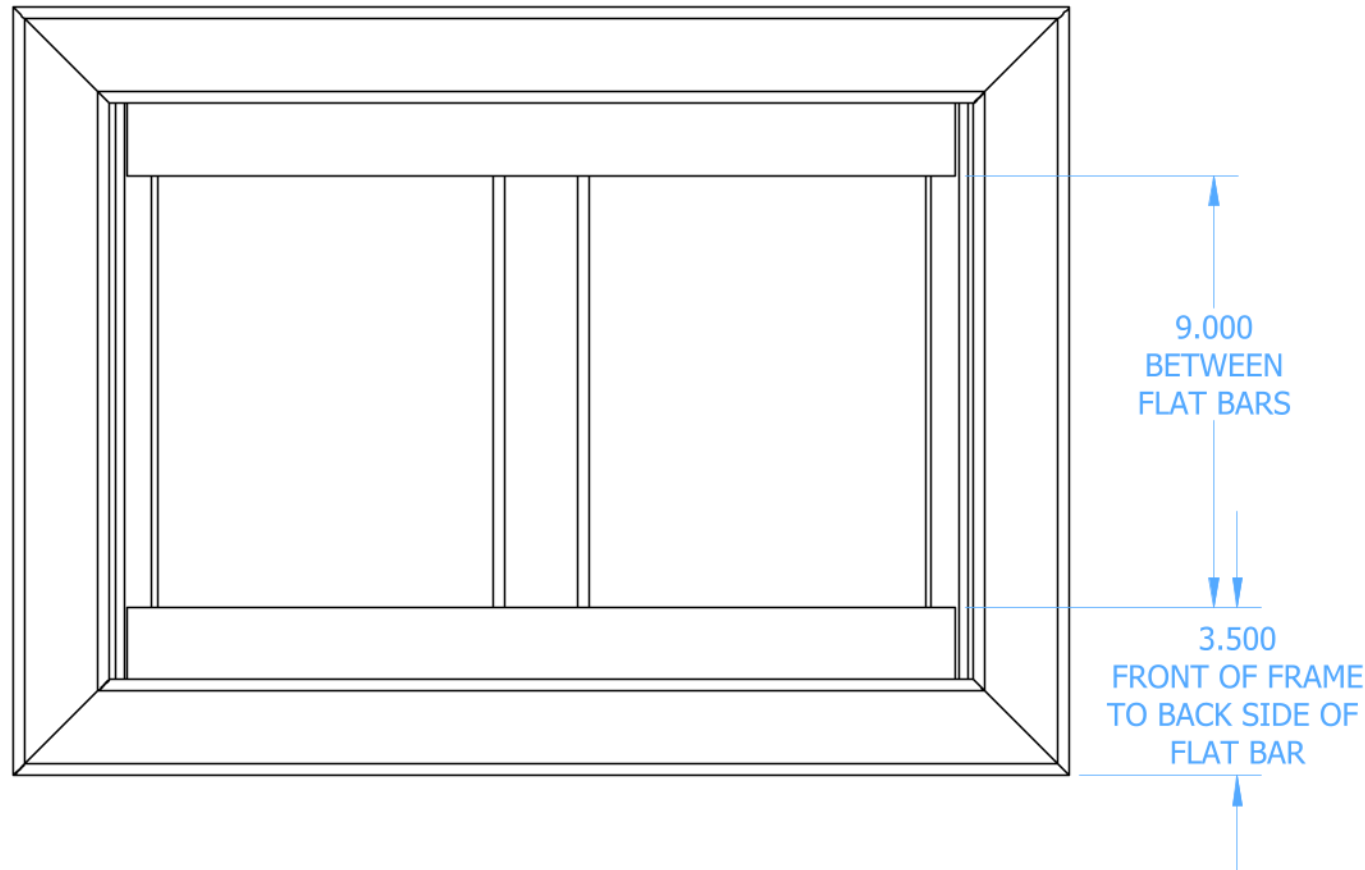
MACHINED BY	DATE / TIME	QTY / THIS ORDER - 1
INSPECTED BY	DATE / TIME	QTY / ASSY - 1
		MATERIAL -
		FINISH - NA
THIRD ANGLE PROJECTION		NOTES - Assembly
		UNLESS OTHERWISE SPECIFIED ALL DIMENSIONS ARE IN INCHES
		TOLERANCES
		DECIMALS .XX ±.01 .XXX ±.005 .XXXX ±.0005
		ANGLES ±1° .X ±.5° .XX ±.1°
		FRACTIONS ±1/16

TAG	X LOC	Y LOC	SIZE
A1	.75	41.25	Ø .266 THRU ALL For Ø .266 SCREW CLEARANCES
A2	.75	29.25	
A3	.75	17.25	
A4	.75	5.25	
A5	11.75	45.75	
A6	11.75	.75	
A7	23.75	45.75	
A8	23.75	.75	
A9	25.57	28.69	
A10	25.57	17.81	
A11	26.07	29.56	
A12	26.07	16.94	
A13	26.66	28.07	
A14	26.66	18.43	
A15	27.16	28.93	
A16	27.16	17.57	
A17	35.75	45.75	
A18	35.75	.75	
A19	44.84	28.93	
A20	44.84	17.57	
A21	45.34	28.07	
A22	45.34	18.43	
A23	45.93	29.56	
A24	45.93	16.94	
A25	46.43	28.69	
A26	46.43	17.81	
A27	47.75	45.75	
A28	47.75	.75	
A29	59.75	45.75	
A30	59.75	.75	
A31	71.75	45.75	
A32	71.75	.75	
A33	82.75	41.25	
A34	82.75	29.25	
A35	82.75	17.25	
A36	82.75	5.25	
B1	.75	21.25	Ø .201 THRU 1/4-20 UNC THRU
B2	.75	14.58	
B3	.75	7.92	
B4	.75	1.25	
B5	6.00	22.13	
B6	6.00	11.25	
B7	6.00	.38	
B8	7.50	22.13	
B9	7.50	11.25	
B10	7.50	.38	
B11	16.25	21.25	
B12	16.25	14.58	
B13	16.25	7.92	
B14	16.25	1.25	Ø 2.000 THRU
C1	3.25	43.25	
C2	36.00	13.25	
C3	51.48	41.60	
C4	55.98	11.07	Ø .397 THRU ALL For Ø .397 SCREW CLEARANCES
D1	4.85	31.36	
D2	5.00	43.50	
D3	5.00	33.50	
D4	5.15	29.64	
D5	15.00	43.50	
D6	15.00	33.50	
D7	20.50	28.50	
D8	20.50	18.00	
D9	24.50	26.75	
D10	24.50	19.75	
D11	26.50	26.75	
D12	26.50	19.75	
D13	59.50	42.75	
D14	59.50	27.75	
D15	60.50	24.75	
D16	60.50	7.75	
D17	70.00	42.75	
D18	70.00	27.75	
D19	70.50	24.75	
D20	70.50	7.75	
D21	80.50	42.75	
D22	80.50	27.75	
D23	80.50	24.75	
D24	80.50	7.75	Ø 4.000 THRU
E1	10.00	38.50	
F1	24.00	41.50	Ø .354 THRU ALL For M8 SCREW CLEARANCES
F2	24.00	37.17	
F3	28.33	41.50	
F4	28.33	37.17	
F5	35.17	6.83	
F6	35.17	2.50	
F7	39.50	6.83	
F8	39.50	2.50	
G1	24.35	41.66	Ø .236 THRU ALL For Ø6.0 DOWEL HOLES SLIP FIT
G2	27.98	37.01	
G3	33.64	20.38	
G4	35.52	6.99	
G5	38.36	26.12	
G6	39.15	2.34	
H1	25.50	23.25	Ø 1.250 THRU
J1	33.24	26.01	
J2	33.24	20.49	Ø .260 THRU ALL For M6 SCREW CLEARANCES
J3	38.76	26.01	
J4	38.76	20.49	
K1	36.00	23.25	
N1	53.48	33.60	Ø .201 THRU ALL 1/4-20 UNC THRU ALL
N2	53.48	31.60	
N3	53.48	29.60	
N4	54.48	23.07	
N5	54.48	21.07	
N6	54.48	19.07	
N7	56.48	33.60	
N8	56.48	31.60	
N9	56.48	29.60	
N10	57.48	23.07	
N11	57.48	21.07	
N12	57.48	19.07	
N13	57.50	8.50	
N14	57.50	3.00	

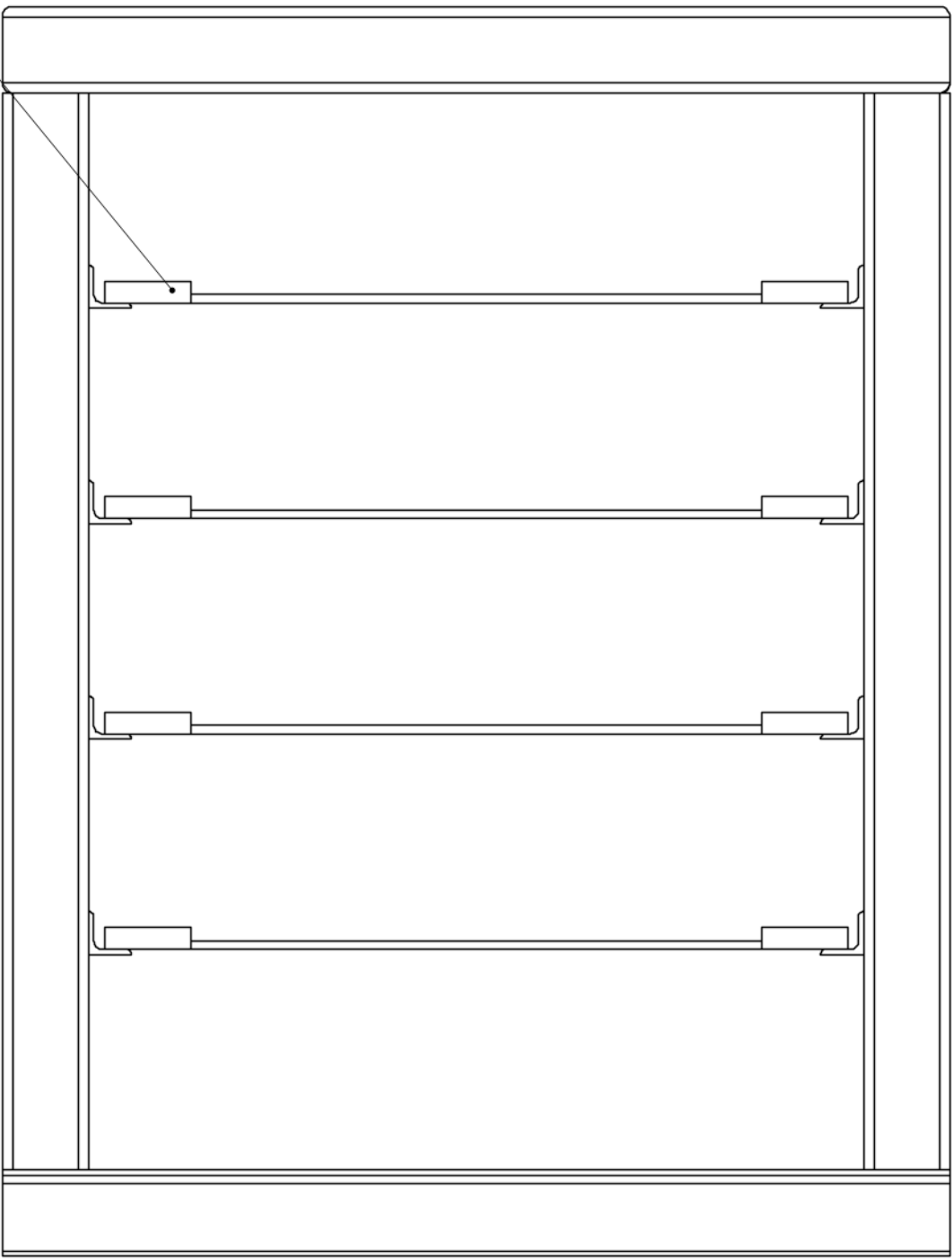


MACHINED BY	DATE / TIME	QTY / THIS ORDER - 1	
INSPECTED BY	DATE / TIME	QTY / ASSY - 1	
		MATERIAL - Aluminum 6061 Alloy	
		FINISH - Powder Coat Grey; Traffic (RAL7042)	
THIRD ANGLE PROJECTION		NOTES - Machined Part	
		UNLESS OTHERWISE SPECIFIED ALL DIMENSIONS ARE IN INCHES	
		TOLERANCES	
		DECIMALS .XX ±.01 .XXX ±.005 .XXXX ±.0005	
		ANGLES ±1° .X ±.5° .XXX ±.1°	
		FRACTIONS 1/16	

ITEM NO.	QTY.	Description	LENGTH	ANGLE1	ANGLE2	Notes
1	2	TUBE, SQUARE 2.00 x 2.00 x .12 Wall	22	45°	45°	
2	2	TUBE, SQUARE 2.00 x 2.00 x .12 Wall	16	45°	45°	
3	4	TUBE, SQUARE 2.00 x 2.00 x .12 Wall	25	0°	0°	
6	1	TUBE, SQUARE 2.00 x 2.00 x .12 Wall	12	0°	0°	
5	1	Angle 2.00 X 2.00 X .125	22	0°	0°	
6	8	Angle 1.00 X 1.00 X .125	16	0°	0°	
7	8	Flat Bar .19 X 1.50	17.25	0°	0°	
8	1	0118 02 1010-01 Angel Mount	22	0°	0°	
9	16	0118 02 1010-02 Controler Stop				



ALL 4 RACKS ARE THE SAME FRONT AND BACK

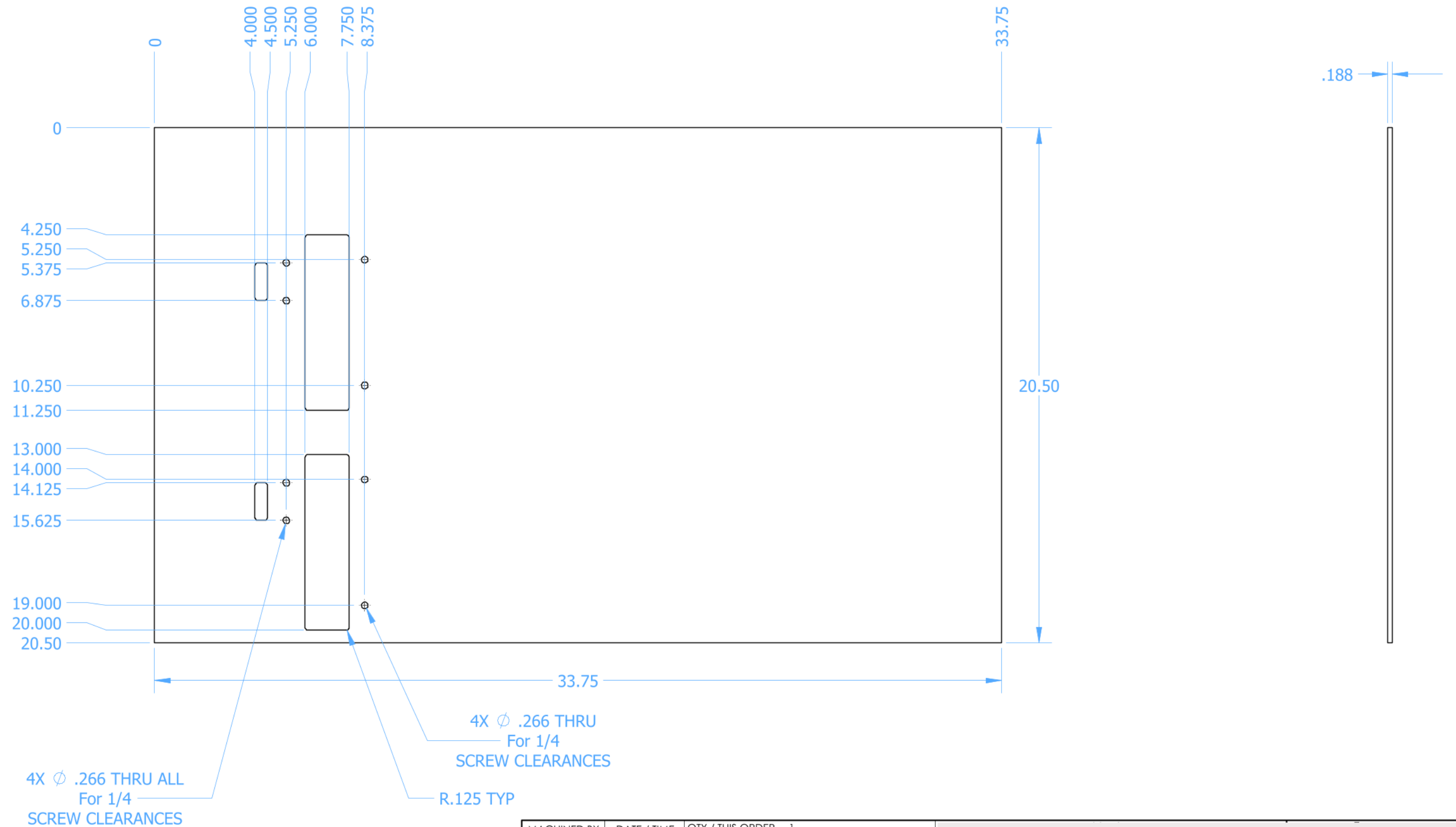


SMALL PLATES
ALIGNED WITH
END OF FLAT BAR

NOTE HOLE LOCATION

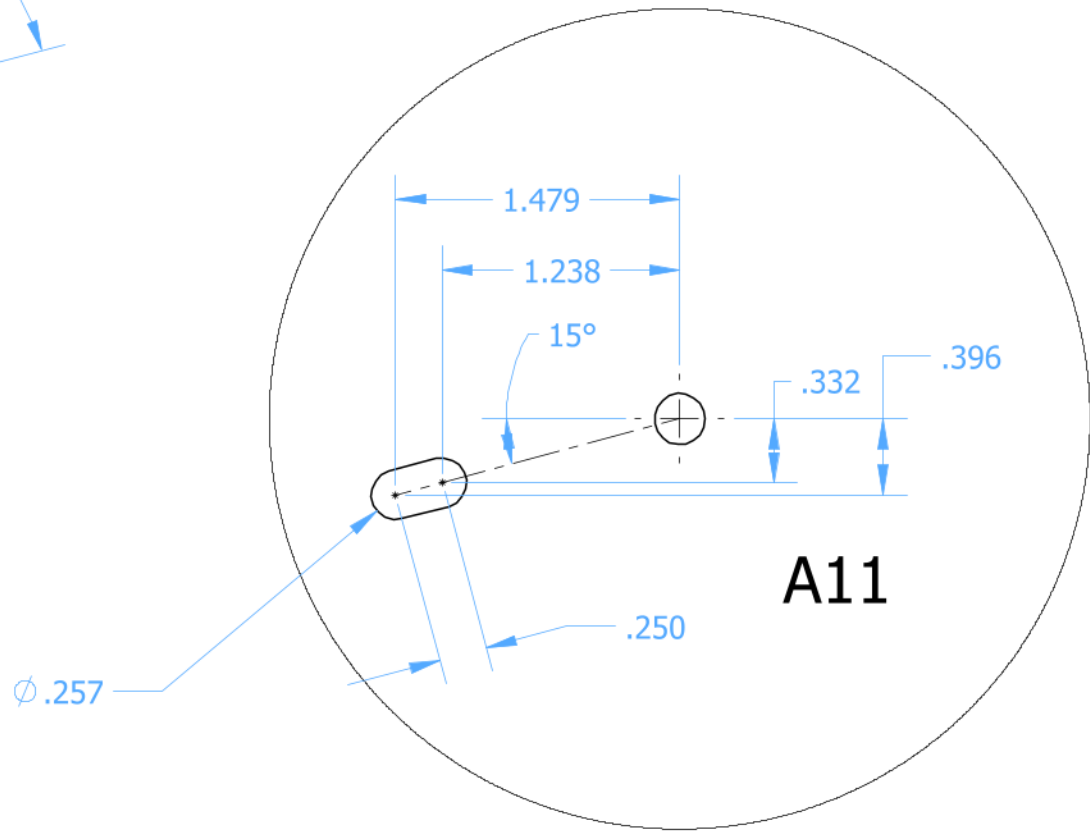
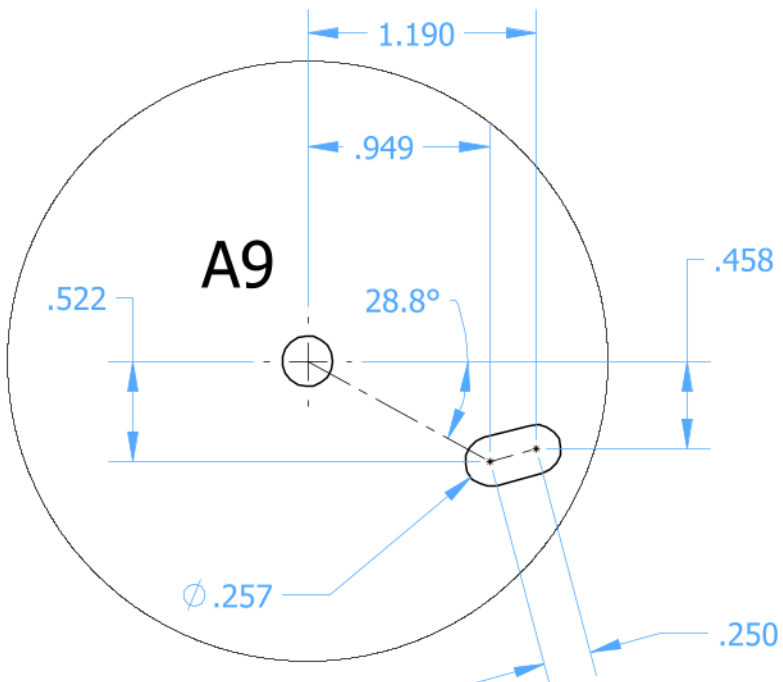
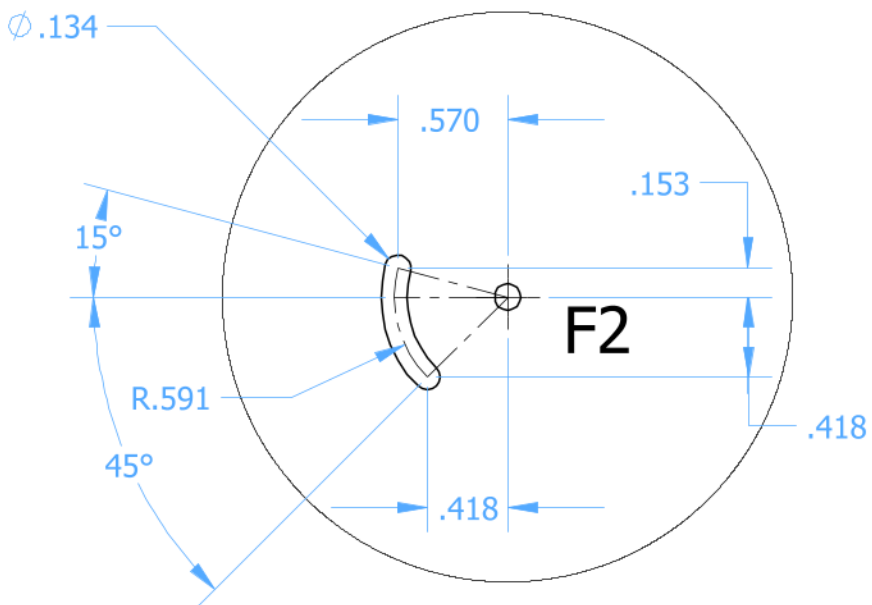
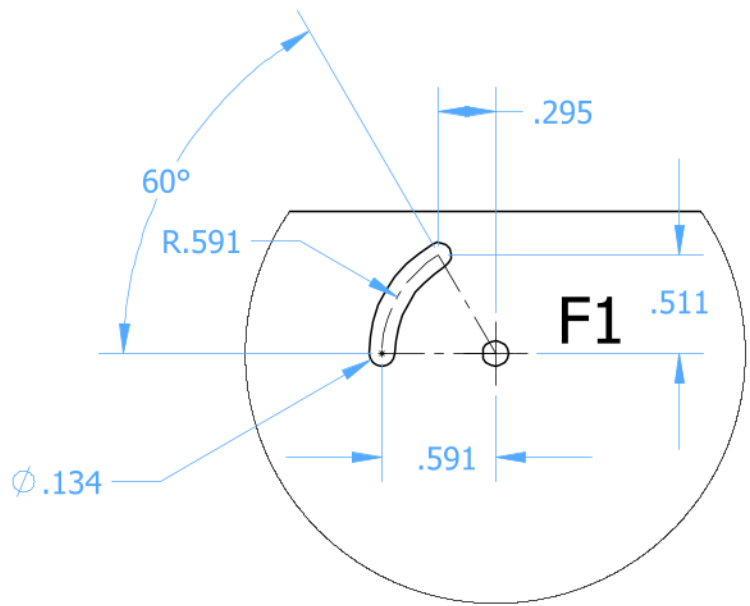
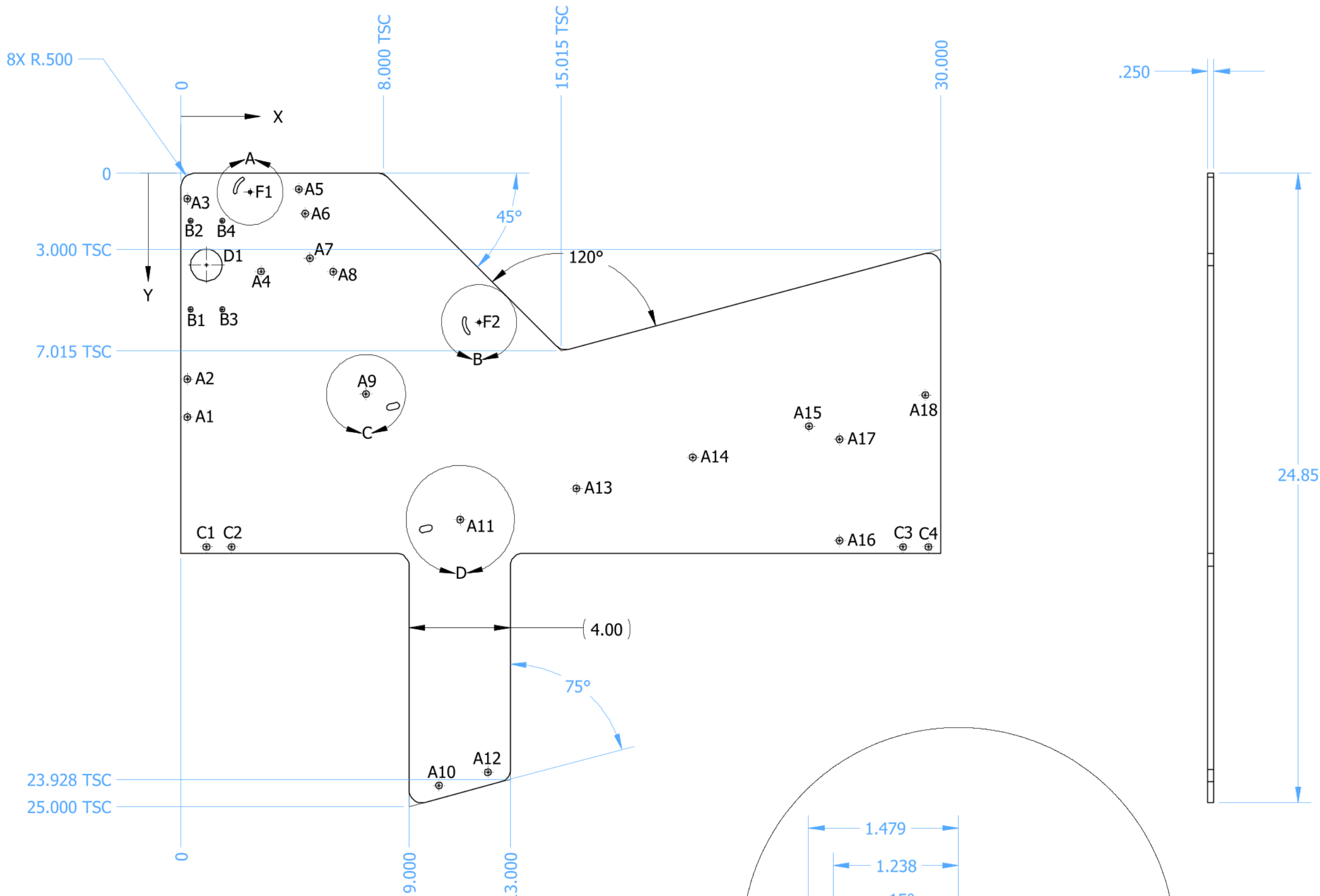
WELD ALL JOINTS
GRIND EXTERIOR SURFACES SMOOTH

MACHINED BY	DATE / TIME	QTY / THIS ORDER - 1
INSPECTED BY	DATE / TIME	QTY / ASSY - 1
THIRD ANGLE PROJECTION		MATERIAL - Plain Carbon Steel
		FINISH - Powder Coat Grey; Traffic (RAL-7042)
		NOTES - Weldment
		UNLESS OTHERWISE SPECIFIED ALL DIMENSIONS ARE IN INCHES
		TOLERANCES
		DECIMALS .XX ±.01 .XXX ±.005 .XXXX ±.0005
		ANGLES ±1° 'X ±.5° 'XX ±.1°
		FRACTIONS ±1/16



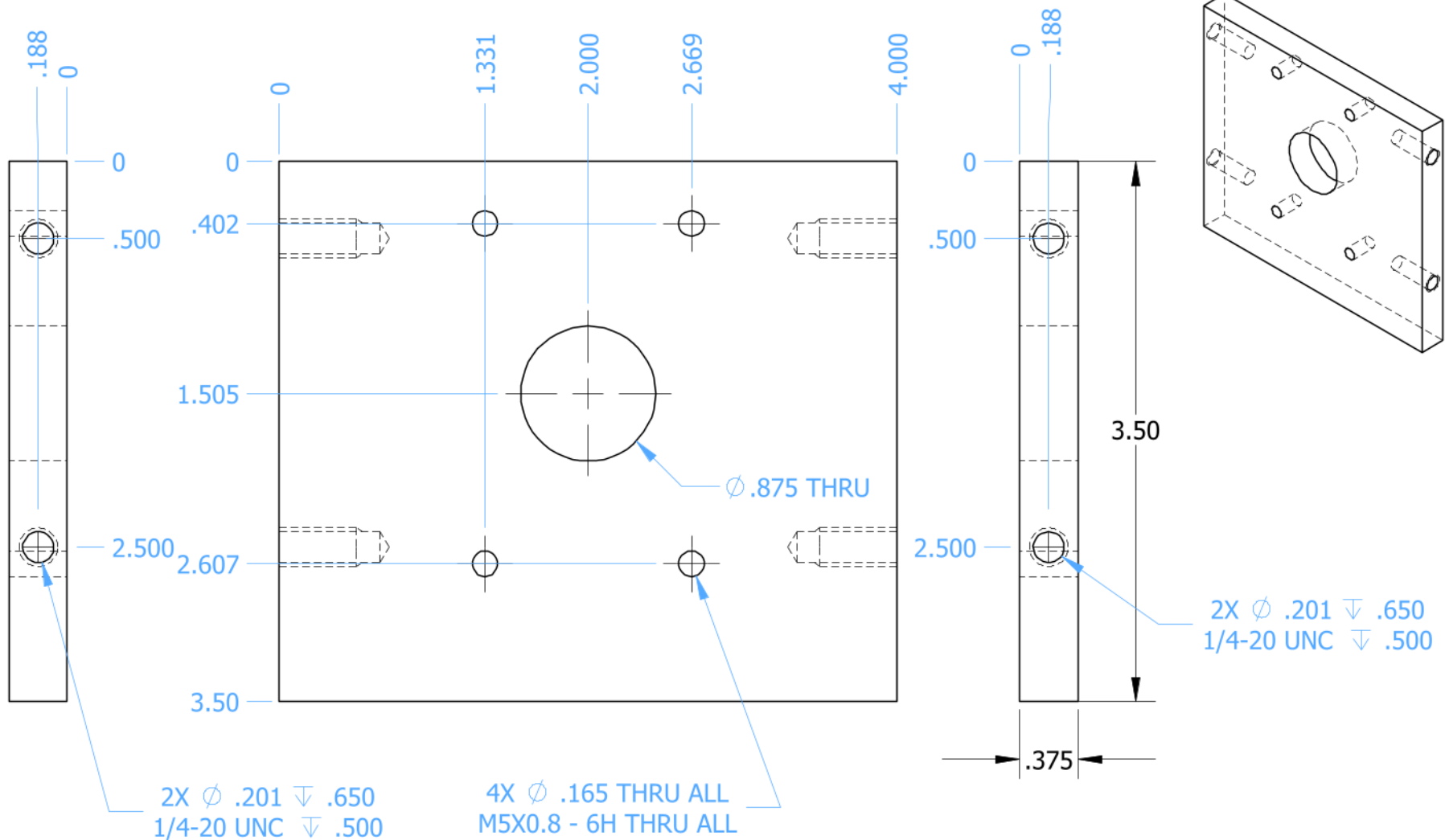
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		QTY / ASSY - 1	
INSPECTED BY	DATE / TIME	MATERIAL - Lexan	
		FINISH - Natural	
THIRD ANGLE PROJECTION		NOTES - Machined Part	
			
UNLESS OTHERWISE SPECIFIED ALL DIMENSIONS ARE IN INCHES			
TOLERANCES DECIMALS .XX ±.01 .XXX ±.005 .XXXX ±.0005 ANGLES ±1° .X ±.5° .XX ±.1° FRACTIONS ±1/16			

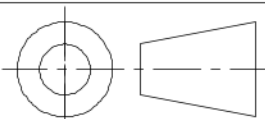
TAG	X LOC	Y LOC	SIZE
A1	.25	9.63	Ø .266 THRU ALL SCREW CLEARANCES
A2	.25	8.13	
A3	.25	1.00	
A4	3.16	3.87	
A5	4.65	.62	
A6	4.91	1.59	
A7	5.09	3.35	
A8	6.01	3.88	
A9	7.30	8.71	
A10	10.18	24.17	
A11	11.02	13.67	Ø .201 THRU ALL 1/4-20 UNC THRU ALL
A12	12.12	23.65	
A13	15.61	12.44	
A14	20.20	11.21	
A15	24.79	9.98	Ø .266 ▽ .650
A16	26.00	14.50	
A17	26.00	10.50	
A18	29.38	8.75	
B1	.38	5.38	Ø 1.250 THRU
B2	.38	1.88	
B3	1.63	5.38	
B4	1.63	1.88	
C1	1.00	14.75	Ø .134 THRU ALL For M3 SCREW CLEARANCES
C2	2.00	14.75	
C3	28.50	14.75	
C4	29.50	14.75	
D1	1.00	3.63	
F1	2.72	.74	
F2	11.77	5.88	

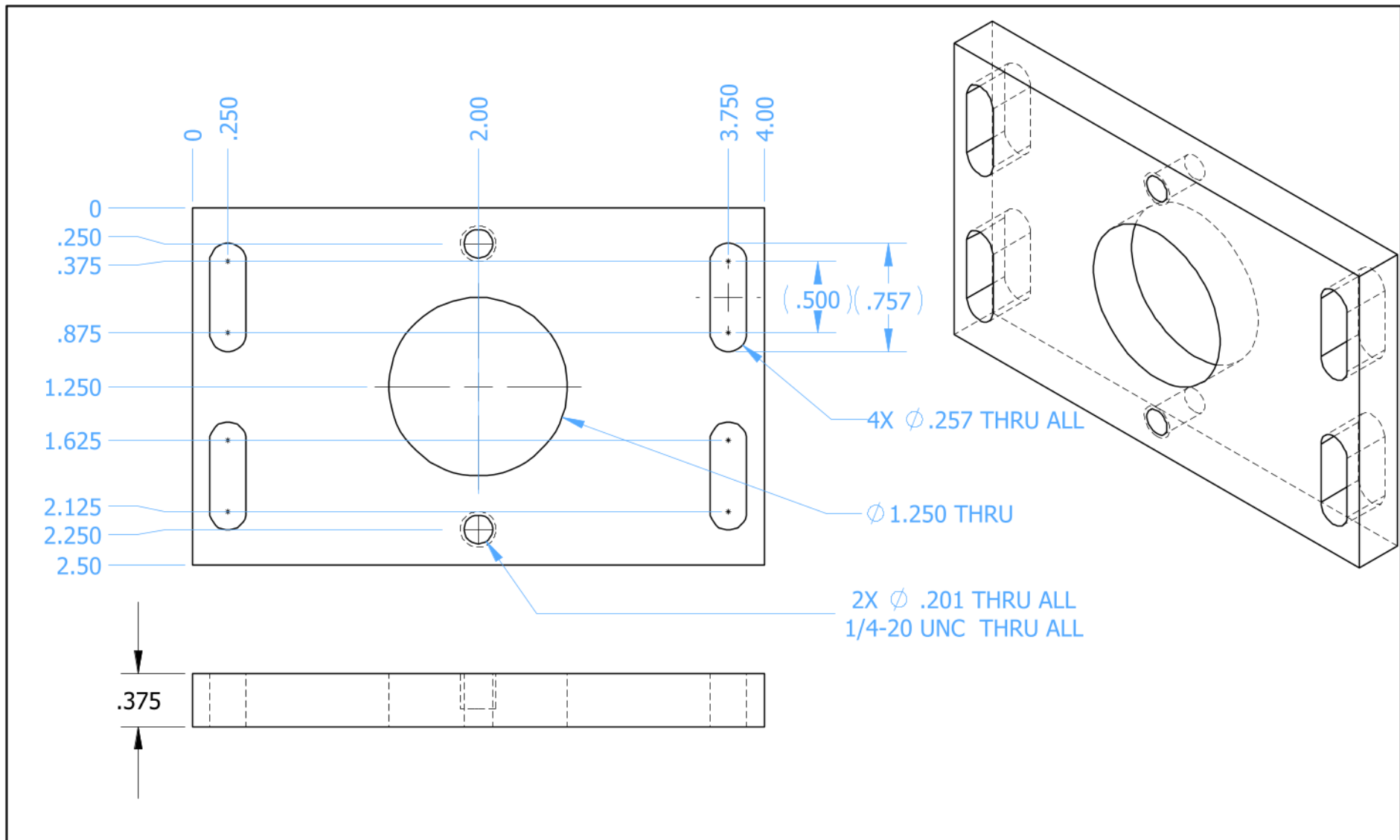


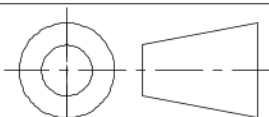
TSC: THEORETICAL SHARP CORNER

MACHINED BY	DATE / TIME	QTY / THIS ORDER - 3
INSPECTED BY	DATE / TIME	QTY / ASSY - 3
		MATERIAL - Aluminum 6061 Alloy
		FINISH - Anodize Clear
		NOTES - Machined Part
THIRD ANGLE PROJECTION		UNLESS OTHERWISE SPECIFIED ALL DIMENSIONS ARE IN INCHES
		TOLERANCES
		DECIMALS .XX ±.01 .XXX ±.005 .XXXX ±.0005
		ANGLES ±1° .X ±.5° .XX ±.1°
		FRACTIONS ±1/16



MACHINED BY	DATE / TIME	QTY / THIS ORDER - 2						
		QTY / ASSY - 2						
INSPECTED BY	DATE / TIME	MATERIAL - Aluminum 6061 Alloy						
		FINISH - Anodize Clear						
THIRD ANGLE PROJECTION		NOTES - Machined Part						
		UNLESS OTHERWISE SPECIFIED ALL DIMENSIONS ARE IN INCHES						
		TOLERANCES	SIZE	DATE CREATED:		REV.		
		DECIMALS .XX ±.01 .XXX ±.005 .XXXX ±.0005	A	DATE PRINTED:				
		ANGLES ±1° .X ±.5° .XX ±.1°		SCALE: 1:1				
		FRACTIONS ±1/16			SHEET 1 of 1			



MACHINED BY	DATE / TIME	QTY / THIS ORDER - 3						
		QTY / ASSY - 3						
INSPECTED BY	DATE / TIME	MATERIAL - Aluminum 6061 Alloy						
		FINISH - Anodize Clear						
THIRD ANGLE PROJECTION		NOTES - Machined Part						
		UNLESS OTHERWISE SPECIFIED ALL DIMENSIONS ARE IN INCHES						
		TOLERANCES		SIZE			REV.	
		DECIMALS .XX ±.01 .XXX ±.005 .XXXX ±.0005		A				
		ANGLES ±1° .X ±.5° .XX ±.1°						
		FRACTIONS ±1/16						
				SCALE: 1:1		SHEET 1 of 1		

.25

Ø 3.00

.331

0

.375

0

.375

.800

Ø .157 THRU ALL
For Ø4.0
DOWEL HOLES
PRESS FIT

22.50°

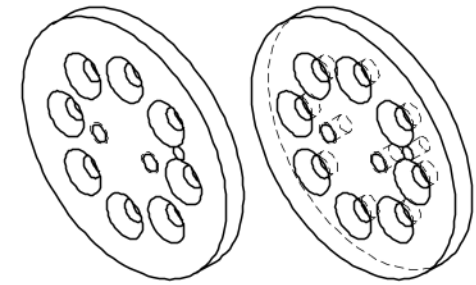
2X Ø .201 THRU ALL
1/4-20 UNC THRU ALL
TAPPED HOLE

8X Ø .260 THRU ALL
✓ Ø .526 X 90°

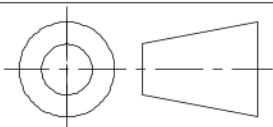
For M6
FLAT HEAD SCREW - ANSI B18.6.7M
Flush or Below Surface

Ø 1.732

45°



MACHINED BY	DATE / TIME	QTY / THIS ORDER - 1
		QTY / ASSY - 1
INSPECTED BY	DATE / TIME	MATERIAL - Aluminum 6061 Alloy
		FINISH - Anodize Clear
THIRD ANGLE PROJECTION		NOTES - Machined Part



UNLESS OTHERWISE SPECIFIED ALL DIMENSIONS ARE IN INCHES			
TOLERANCES			
DECIMALS	.XX ±.01	.XXX ±.005	.XXXX ±.0005
ANGLES	±1°	.X ±.5°	.XX ±.1°
FRACTIONS	±1/16		

SIZE
A

SCALE: 1:1

SHEET 1 of 1

REV.